The Cost of Asking: A Fine-Grained Taxonomy of Hallucination in LLM-Generated Islamic Citations

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Abstract

Large Language Models (LLMs) are increasingly consulted for Islamic questions, where answers rest on evidence from the Quran and the Hadith. However, existing hallucination labels conflate a minor phrasing alteration with a complete fabrication or a misattribution. We introduce a fine-grained taxonomy of Islamic citation errors, derived from expert adjudication notes, that separates textual fidelity (correct, corrupted, fabricated) from attributional accuracy (narrator chain and source). With an LLM annotator validated against three domain experts, we benchmark twelve LLMs under explicit and implicit citation prompts and eight more under the implicit prompt. Most open-weight models fail mainly by fabrication, while frontier models almost never fabricate but alter wording and, unprompted, leave more than half of their quotations unsourced. Asking for citations costs open-weight models 14.2 points of exact-text accuracy. The citations they would volunteer anyway do not get worse; the loss falls on the additional citations the instruction forces, which are more often fabricated, and the sources added to quotations they would otherwise leave unsourced are wrong as often as right. Frontier models lose no accuracy, and asking them to cite turns their unsourced quotations into correctly sourced ones. We release the fine-grained annotations and guidelines.

1 Introduction

LLMs have become an important tool for question answering, but are prone to hallucination, which is described as generated text that seems plausible, yet is factually incorrect or unfaithful (Huang et al., 2025). In the Islamic domain, answering religious questions fundamentally requires authoritative evidence from scripture, namely the Quran, which Muslims acknowledge as the divine text revealed in Arabic, and the Hadith, which document the teachings and practices of the Prophet Muhammad (Bashir et al., 2022). Because LLM responses rely directly on these texts, errors in quoting or citing them can carry ethical, theological, and social consequences and potentially help spread misinformation (Mubarak et al., 2025; Fawzi et al., 2026). A religious citation consists of a quoted text and its attribution. A Quranic citation quotes a verse of the Quran (Ayah) and attributes it to a chapter(Surah) and verse number. A Hadith citation quotes the text of the report (Matn) along with the chain of narrators who transmitted it (Isnad) and the collection that records it. Each field can be correct or incorrect independently. A model may quote a verse correctly while assigning it to the wrong chapter, or reproduce an accurate Hadith report while naming the wrong narrator. A binary label cannot pinpoint such errors or separate errors of different severity. Therefore, a phrasing issue and a misattributed source receive the same label. A recent survey of AI systems for Islamic knowledge finds that models produce fluent answers whose cited support is missing, misattributed, or does not actually back the claim (Bhatia et al., 2026a).

In this paper, we propose a taxonomy to evaluate each citation along two axes, textual fidelity and attributional accuracy. Textual fidelity examines whether the quoted Ayah or Matn reproduces an attested text, distinguishing exact matches, recoverable corruptions, and ungrounded fabrications. Attributional accuracy determines whether the Isnad and source point to the correct origin. This reveals the errors models actually make when citing religious text, moving beyond theoretically possible errors. Furthermore, we conduct the evaluation entirely in Arabic, the language of the source texts, which also makes the task challenging because Arabic is morphologically rich, and a single character or diacritic change can alter a verse's meaning. To apply the framework at scale, an LLM annotator labels every extracted citation, validated against a reference built from three domain experts. We apply the framework to twelve models under the two

ways users actually encounter citations. Explicit instruction to cite evidence and an implicit condition where models volunteer citations unprompted. We extend the implicit benchmark with eight additional recent open-weight models. Comparing the two conditions answers a practical question. Islamic assistants and their system prompts routinely instruct models to support every answer with Quran and Hadith. Does demanding citations make the citations worse? We decompose the accuracy change between the two conditions into three parts: which questions receive citations, the quality of the citations a model would volunteer anyway, and the citations the instruction adds.

Our contributions are:

- We introduce a fine-grained Islamic citation evaluation taxonomy that decouples textual fidelity from attributional accuracy, with error categories derived from and validated against expert human consensus.
- We benchmark twenty LLMs (Arabic-centric, multilingual, and frontier) on fine-grained annotations across explicit and implicit citation prompts.
- We analyse how instructing models explicitly to cite changes citation accuracy, comparing demanded citations against those models volunteer unprompted.

2 Background and Related Work

Islamic Citations The Quran, the holy book of Muslims, is a fixed corpus of 114 chapters (*Surahs*) and 6,236 verses (*Ayahs*), each identified by chapter and verse number (Abokhodair et al., 2020). A Hadith is any speech, action, approval, or description attributed to the Prophet Muhammad (Fawzi et al., 2025), comprising a chain of narrators (*Isnad*) tracing its transmission and the reported text itself (*Matn*) (Azmi et al., 2019). Hadiths are compiled in collections, most authoritatively the six canonical collections (*al-Kutub al-Sitta*), and Islamic scholars grade their authenticity as authentic, good, weak, or forged, based on both the chain and the text (Fawzi et al., 2026). Two properties separate Hadith from the Quran. First, Hadith lacks a single canonical wording: the same report is transmitted through several chains with different wording, and transmission by meaning (*riwaya bi-l-ma'na*) is accepted (Fawzi et al., 2026), so a criterion that accepts only one collection's wording misclassifies genuine transmissions. Second, attribution is multi-layered: a citation may name a narrator, a collection, a grading, and a number, and each can be independently right or wrong while the *Matn* is exact. We evaluate citations rather than authenticity. Because a report classified as weak is still historically documented as a Hadith variant, a model that reproduces it faithfully is labeled correct on the textual axis.

Fine-grained Hallucination Detection Binary labels do not identify the affected span, distinguish partial hallucination, or represent severity (Alansari and Luqman, 2026); fine-grained methods decompose outputs into task-suited units: spans (Mishra et al., 2024), atomic claims (Wan et al., 2024; Zhang et al., 2024), entities and relations (Abdaljalil et al., 2025), or reasoning steps (Zhang et al., 2025; Li et al., 2025). In Arabic, AraHalluEval (Alansari and Luqman, 2025) applies 12 fine-grained indicators at the claim and sentence level, and ArabiGEE (Elhady et al., 2026) labels grammatical errors at the character- and word-span level through a hierarchical taxonomy. These studies show that evaluation granularity and taxonomy should follow the task being evaluated; we apply the same principle by deriving our taxonomy from errors observed in individual components of Quranic and Hadith citations.

LLM Evaluation in the Islamic Domain Existing work spans knowledge benchmarks with Multiple Choice Questions and answer spaces (Abdelaal et al., 2026; Lahmar et al., 2025; Bouchekif et al., 2025), retrieval-based mitigation systems evaluated at the response level (Mohammed et al., 2025; Al-Azani et al., 2025; El Ganadi et al., 2025), and detection frameworks over free-form output (Bhatia et al., 2026b; Elmahjub et al., 2026; Mushtaq et al., 2025). The closest prior work is the IslamicEval benchmarks: IslamicEval 2025 (Mubarak et al., 2025) first moved detection to individual Quranic and Hadith spans, with a single binary label per quotation and no evaluation of attribution. IslamicEval 2026 (Alharbi et al., 2026) extends this to the segment level over a substantially larger corpus, labelling the Isnad and claimed source only when the quoted text is correct, with every label being binary.

3 A Taxonomy of Islamic Citation Errors

We developed our taxonomy empirically, from the distinctions three domain experts drew in their free-text adjudication notes while annotating a sample of model-generated citations (Section 4). The taxonomy, summarised in Figure 1, evaluates every citation along two axes.

Textual Fidelity measures how accurately the quoted Ayah or Hadith Matn reproduces the text it claims to quote, with three labels:

- **Correct**: an exact verbatim reproduction of an attested text.
- **Corrupted**: a real text underlies the citation but the wording has been altered; the intended text is recoverable.
- **Fabricated**: no attested text underlies the citation; it cannot be traced to the Quran or to any documented Hadith collection.

Recoverability is what separates the two error labels: a corrupted citation misrepresents a text that exists, while a fabricated one invents scripture.

Attributional Accuracy measures whether the citation's attribution points to the correct origin, with a binary label per component. A Hadith citation carries two attribution components, judged independently: the Isnad and the claimed source collection. A Quranic citation carries one: the claimed chapter and verse. The two axes are connected through the quoted text: attribution can only be judged for a text that exists. When the citation is correct or corrupted, the Isnad and claimed source are labelled against the cited text; when it is fabricated, attribution is not judged. Unsupplied components are not penalised; citations providing no attribution form separate unsourced modes. Figure 1 sets out the assignment procedure with full failure mode definitions, examples, and expert annotations detailed in Appendix B.

We use this taxonomy to benchmark citation generation: we assign each extracted citation in both conditions a failure mode using the procedure above, and Section 5 reports the labels and analyses the two sets.

4 Data and Annotation

Questions and Conditions The benchmark covers 500 real user questions requiring religious citations, drawn from the interaction logs of Fanar, a deployed Arabic assistant (Team et al., 2026). Twelve models answered these questions, including ten open-weight (3B–13B) and two frontier (GPT-5.4, Gemini 3.1 Pro) models. Each model answered every question under two conditions, all in Arabic. In the explicit condition, the instruction was to answer the question and cite religious evidence, including its source. In the implicit condition, the models were instructed to answer with no mention of citations. We cleaned both conditions identically, excluding responses that were degenerate, heavily code-switched, truncated, or empty. All other settings were identical. We provide both prompts in Appendix A.1. In decoding, we used each model's intended configuration where the model ships one, and otherwise set temperature to 0.7 with nucleus top-p 0.9. We obtained the explicit-condition responses, their extracted citations, and the expert annotations, including adjudication notes, with permission from the authors of (Alharbi et al., 2026). While they released binary labels, we then re-annotated every citation in both conditions under the fine-grained taxonomy in Section 3. An extended panel of eight recent open-weight models (9B–40B) broadens the implicit-condition benchmark to current releases and larger scales. These models were selected from Arabic LLM leaderboards[1] within a 40B compute budget. All cross-condition comparisons use the matched twelve. Table 1 summarises the retained responses and citations of two conditions after cleaning.

	Matched (12)		Extended (8)
	Explicit	Implicit	Implicit
Responses annotated	4,537	5,845	3,814
open-weight	3,706	4,845	3,814
frontier	831	1,000	—
With ≥1 citation	4,535	1,442	1,265
Citations	16,529	2,827	2,514

Table 1: Responses and citations per condition over the 500 shared questions. The matched twelve models were run under both conditions; the extended panel of eight recent open-weight models was run under the implicit condition only. The explicit condition is restricted to the publicly released train and dev splits; the implicit set was constructed and used fully.

Expert Reference We randomly selected citations (250 Hadith, 100 Quranic) that were indepen-

[1] https://huggingface.co/spaces/OALL/Open-Arabic-LLM-Leaderboard; https://crfm.stanford.edu/helm/arabic/latest/

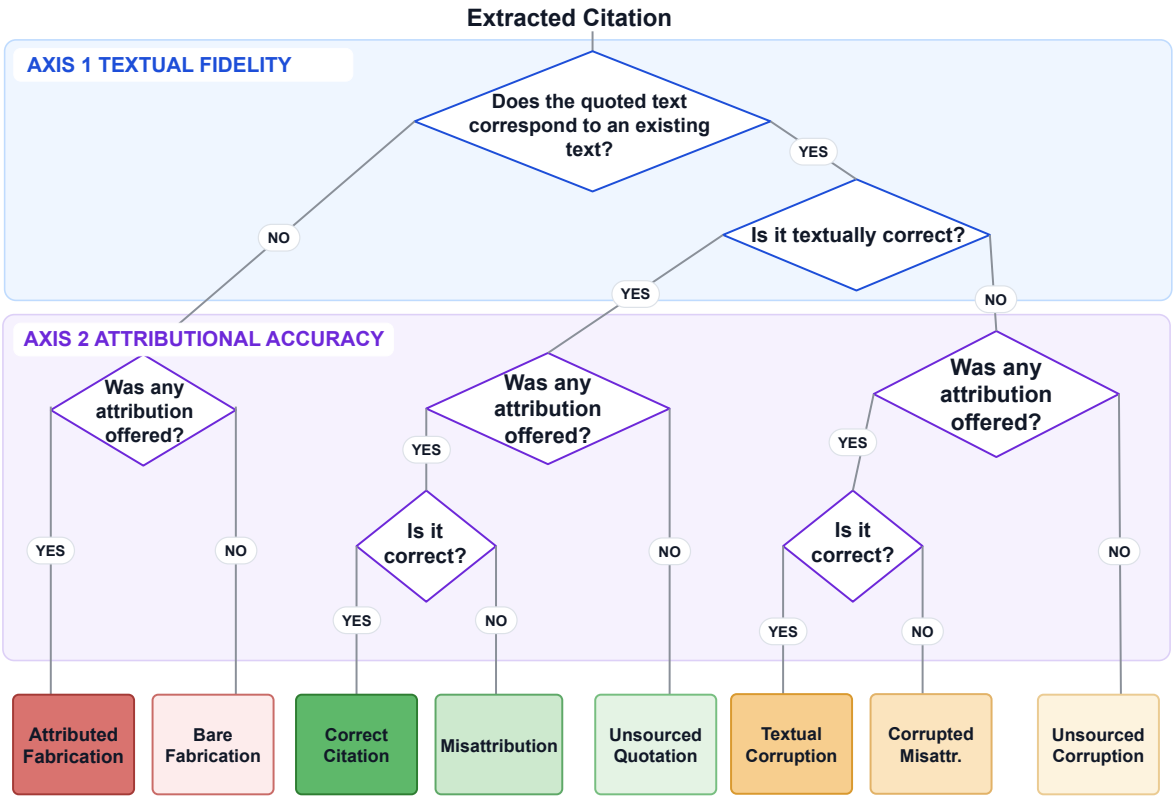


Figure 1: Overview of the fine-grained taxonomy as a decision procedure. Three ordered questions, does the quoted text correspond to an existing text, is it textually correct, and was an attribution offered and is it correct, assign every citation to one of eight modes, from fully correct citation to bare fabrication. Colours mark the textual-fidelity class.

dently annotated in Arabic by three Islamic domain experts. Each expert has 20–30 years of experience in editing and authentication of Islamic heritage manuscripts. They labelled each component and added free-text notes. Expert agreement (Fleiss' κ) ranges from almost perfect on the Quranic claimed source (0.870) and Hadith Isnad (0.829) to moderate on the Hadith Matn (0.538) and its claimed source (0.580), with the Ayah substantial (0.691). The annotators recognised the cited Hadith but applied different exactness thresholds. Furthermore, because the same report appears in multiple narrations across a vast corpus, they sometimes judged the wording and its source against different attested variants. We derived fine-grained ground truth from the notes. A correction naming the cited text yields *corrupted*, while a statement that the citation has no basis (*la aṣla lah*) yields *fabricated*. Correct judgments carried over unchanged. We excluded citations with absent notes, retaining 230/250 Hadith and 91/100 Quranic citations. Appendix B shows example notes and the labels derived from them.

Annotator Validation An annotator model Gemini Pro 3.1 (Gemini Team et al., 2023) labels each component under the rules above, using few-shot English prompts (Gilardi et al., 2023; Tan et al., 2024; Hasanaath et al., 2025; Gupta et al., 2025); against the expert reference it reaches $\kappa = 0.725$ on the Matn and 0.803 on the Ayah over the full three-label space, exceeding the experts' agreement among themselves. We also test the annotator's self-consistency under repeated sampling (Manakul et al., 2023). Across three runs at temperature 1, 92.1% of gold citations received identical labels (mean pairwise $\kappa = 0.91$), with every flip adjacent and concentrated on human-contested rows. A second annotator model (GPT-5.5), run on the same citations, was weaker on the quoted text; requiring the two to agree raised Matn agreement from $\kappa = 0.725$ to 0.791 but left 15% of citations unlabelled, so Gemini Pro 3.1 alone annotated all citations. To validate the fidelity split, an independent correction check attempted to recover each citation's intended wording from the canonical corpora, across all 16,529 explicit-condition citations:

88.2% of *corrupted* citations received a grounded correction, while 79.0% of *fabricated* ones were marked not correctable.

5 Results and Analysis

5.1 Fine-Grained Benchmark

Overall Performance Table 2 gives the distribution of citations over the eight modes when models answer without being asked to cite; the explicit-condition distribution is given in Appendix 7. The frontier models quote the text correctly in 90.9% to 91.6% of their citations. While, the matched open-weight models range from 9.4% to 80.1% (pooled rate: 39.9%), and the extended panel ranges from 34.1% to 85.0% (pooled rate: 57.5%).

The two error labels separate the models further. The frontier models produce one fabricated citation, so nearly all of their errors are corruptions of real text. Among the open-weight models, the split follows accuracy: the seven most accurate, from Gemma-4-31B (85.0%) down to Qwen3.5-27B (59.1%), all corrupt at least as often than they fabricate, while every model below them fabricates more often than it corrupts. In the matched panel, where only ALLaM-7B and Fanar-1-9B fall in the first group, fabrication makes up three-quarters of the pooled errors (45.0 of 60.1 points), and eight of the ten models produce more fabricated citations than correct ones. A binary label would count all of these errors the same, although only the corruptions point back to a text that exists.

Attribution The attribution axis adds what the text label alone cannot show. Fabricated citations usually carry a source; in every matched open-weight model, most fabricated citations are attributed, and, pooled, 34.4 of the 45.0 points of fabrication carry a named collection, narrator, or surah. A source is therefore no evidence that the quoted text exists. In the other direction, an exact quotation is not always a correct citation. Pooled, 8.3% of the matched open-weight citations and 6.8% of the extended panel's are exact quotations with an incorrect source or narrator. The frontier models instead leave most quotations unsourced, so their exact quotations are rarely misattributed but often unattributed.

Misattribution and Fabricated Sources In both open-weight panels, most misattributed Quranic citations name a different chapter (Surah) altogether (71.9% and 79.4% of classified cases). Fewer name the correct Surah but provide the wrong verse number (17.4% and 9.5%). The remaining cases name a non-existent Surah or cite a verse number beyond the chapter's actual length. For Hadith citations, most misattributions name a collection that does not actually contain the report (86.7% and 80.2%). The rest overclaim, citing a report as universally agreed upon (*muttafaq 'alayh*) or canonical (*Sahih*) when only one or neither of the two primary authoritative collections actually contains it (11.7% and 19.8%). Furthermore, the specific collection the model names provides no reliable information about whether the quoted text is genuine. A completely fabricated Hadith is credited to the most highly regarded collection, *Sahih al-Bukhari*, just as often as a correct one. This pattern holds across both the implicit (28.0% against 30.2%) and explicit conditions (50.4% against 50.6%).

5.2 The Cost of Asking

Pooled Effect Figure 2 (a) shows the citation-weighted accuracy deltas for the pooled open-weight and frontier models. Pooled accuracy for the open-weight models drops by 14.2 points, while the citation count rises from 1,745 to 11,527 and moves the average from 0.36 citations per answer to 3.11. However, the frontier models increase their accuracy under the explicit prompt by 3.3 points. The drop in accuracy can therefore come from two places, the citations a model was already volunteering becoming worse under the demand, or the additional citations the demand forces being worse than the volunteered ones.

Decomposition To isolate the effect of the explicit instruction, we compare each model's citations across the two conditions on the same questions. Accuracy here is the share of citations whose quoted text is labelled correct, weighted by citation. We decompose the change in accuracy from the implicit to the explicit condition into three components:

Selection: the accuracy change from restricting the comparison to questions where the model provided a citation under both prompts.

Quality: the accuracy difference, on the shared questions, between the first k citations a model gives under each prompt, where k is the number of citations it volunteers under the implicit prompt.

		Correct text				Corrupted text				Fabricated		
Model	**N**	**All**	**Correct**	**Misattr.**	**Unsrc.**	**All**	**Corrupt.**	**Misattr.**	**Unsrc.**	**All**	**Attrib.**	**Bare**
GPT-5.4† (Singh et al., 2026)	347	**91.6%**	38.0%	0.6%	53.0%	**8.4%**	5.2%	0.0%	3.2%	**0.0%**	0.0%	0.0%
Gemini3.1-Pro† (Gemini Team et al., 2023)	735	**90.9%**	38.8%	0.1%	52.0%	**9.0%**	4.9%	0.1%	3.9%	**0.1%**	0.0%	0.1%
ALLaM-7B (Bari et al., 2024)	282	**80.1%**	58.5%	9.2%	12.4%	**13.5%**	7.4%	3.2%	2.8%	**6.4%**	3.5%	2.8%
Fanar-1-9B (Team et al., 2025)	335	**66.9%**	31.0%	14.9%	20.9%	**19.4%**	8.1%	6.0%	5.4%	**13.7%**	9.0%	4.8%
AceGPT-v2-8B (Huang et al., 2023)	60	**36.7%**	8.3%	10.0%	18.3%	**25.0%**	8.3%	6.7%	10.0%	**38.3%**	23.3%	15.0%
Jais-13B(Sengupta et al., 2023)	22	**31.8%**	4.5%	0.0%	27.3%	**27.3%**	4.5%	9.1%	13.6%	**40.9%**	22.7%	18.2%
Llama-3.1-8B(Grattafiori et al., 2024)	118	**29.7%**	3.4%	6.8%	19.5%	**10.2%**	5.1%	5.1%	0.0%	**60.2%**	50.0%	10.2%
Command-R-7B (Alnumay et al., 2025)	131	**26.7%**	9.2%	5.3%	12.2%	**20.6%**	9.2%	6.9%	4.6%	**52.7%**	37.4%	15.3%
Gemma-3-12B (Gemma Team et al., 2025)	204	**23.0%**	7.8%	4.9%	10.3%	**23.5%**	10.8%	7.4%	5.4%	**53.4%**	43.6%	9.8%
Phi-3.5-mini(Abdin et al., 2024)	75	**21.3%**	1.3%	4.0%	16.0%	**4.0%**	0.0%	1.3%	2.7%	**74.7%**	64.0%	10.7%
Qwen3-8B (Yang et al., 2025)	422	**17.8%**	5.2%	7.6%	5.0%	**10.2%**	3.6%	3.3%	3.3%	**72.0%**	55.2%	16.8%
Mistral-7B(Jiang et al., 2023)	96	**9.4%**	2.1%	2.1%	5.2%	**6.2%**	1.0%	1.0%	4.2%	**84.4%**	65.6%	18.8%
Extended panel (implicit prompt only)												
Gemma-4-31B (Gemma Team et al., 2026)	320	**85.0%**	38.1%	1.6%	45.3%	**12.8%**	6.9%	1.2%	4.7%	**2.2%**	0.3%	1.9%
Fanar-2-27B (Team et al., 2026)	305	**73.4%**	44.6%	9.8%	19.0%	**16.7%**	8.9%	2.0%	5.9%	**9.8%**	5.6%	4.3%
Karnak-40B (kar, 2026)	168	**73.2%**	42.9%	6.5%	23.8%	**17.3%**	6.0%	6.0%	5.4%	**9.5%**	6.5%	3.0%
Gemma-4-12B (Gemma Team et al., 2026)	231	**66.7%**	20.3%	4.8%	41.6%	**16.9%**	4.3%	3.0%	9.5%	**16.5%**	6.5%	10.0%
Qwen3.5-27B (Qwen Team, 2026)	467	**59.1%**	35.5%	5.6%	18.0%	**28.1%**	12.4%	7.7%	7.9%	**12.8%**	7.5%	5.4%
Qwen3.5-9B (Qwen Team, 2026)	357	**46.8%**	23.8%	9.2%	13.7%	**20.2%**	8.4%	4.2%	7.6%	**33.1%**	18.5%	14.6%
AceGPT-v2-32B (Huang et al., 2023)	73	**38.4%**	13.7%	2.7%	21.9%	**13.7%**	4.1%	6.8%	2.7%	**47.9%**	21.9%	26.0%
Ministral-3-14B (Liu et al., 2026)	593	**34.1%**	19.6%	8.8%	5.7%	**26.6%**	12.6%	9.4%	4.6%	**39.3%**	33.7%	5.6%
Frontier (2)	1,082	**91.1%**	38.5%	0.3%	52.3%	**8.8%**	5.0%	0.1%	3.7%	**0.1%**	0.0%	0.1%
Open-weight (10)	1,745	**39.9%**	19.0%	8.3%	12.6%	**15.1%**	6.3%	4.6%	4.1%	**45.0%**	34.4%	10.7%
Extended panel (8)	2,514	**57.5%**	30.0%	6.8%	20.8%	**21.1%**	9.3%	5.5%	6.2%	**21.4%**	14.4%	7.0%

Table 2: Per-model distribution over the eight terminal classes of the citation taxonomy under the implicit prompt, as a percentage of each model's N citations. *Unsrc.*: no source offered. A Hadith citation counts as correctly attributed only if every component it supplies is correct. †marks frontier models.

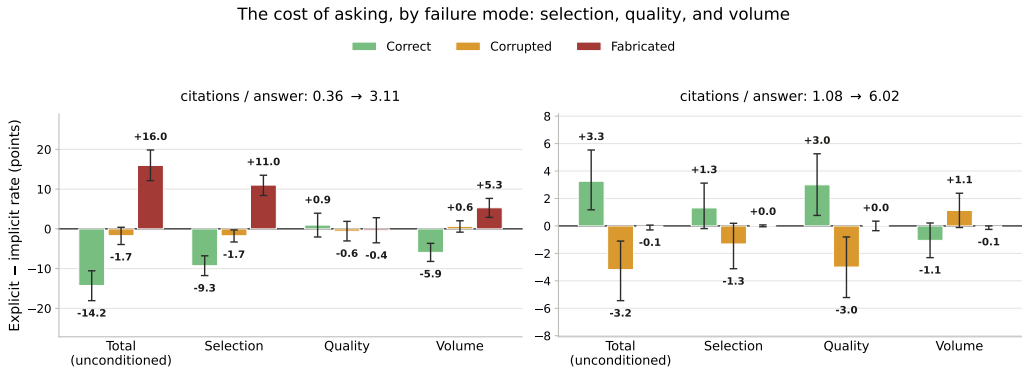
Volume: the accuracy cost of the extra citations the explicit instruction adds beyond what the model volunteers under the implicit prompt.

Selection accounts for most of the accuracy drop, so the largest cost of the instruction comes from asking a model to cite on questions it was not going to cite on under its default behavior. The quality component is indistinguishable from zero. This shows that the citations the model would have volunteered anyway do not become worse by being demanded, which is evident when we also look at only the first citation ($k = 1$) under both conditions, where the difference in pooled accuracy of both settings is nearly zero (46.4% vs 46.5%). The volume component lowers the pooled average by 5.9 points, an interval that excludes zero, where the first shared k citations have an average accuracy of 41.8%, and the remaining citations *beyond* k score 30.5%. Taken together, the three components give a more complete picture: the instruction does not degrade the quality of citations, it exhausts a supply the model has. A model has a few citations per question, and going beyond this cap results in a higher likelihood of hallucination.

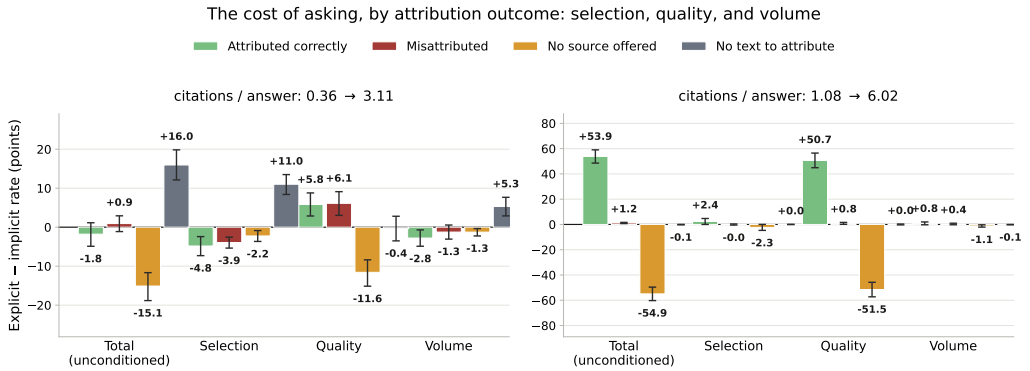
Past k, marginal citations do not merely fail more often; they fail differently. Among the first k explicit citations on shared questions, 41.8% are correct, 14.2% corrupted and 44.0% fabricated. Beyond k, accuracy falls to 30.5% while the corrupted share is barely changes (14.2% to 15.4%) and the fabricated share rises to 54.1%. Suggesting that when pushed past the supply of citations it can ground, a model does not misquote more, it invents more.

Attribution. On the attribution axis (Figure 2b), the decomposition tracks four outcomes: correctly attributed, misattributed, unsourced, and fabricated text. For the open-weight models, the instruction barely moves correct attribution (-1.8) or misattribution ($+0.9$). What changes is that 15.1 points of unsourced quotations are replaced by 16.0 points of fabricated text, almost entirely through selection ($+11.0$) and volume ($+5.3$), the same components that carried the loss in textual fidelity.

Quality is the only component where the instruction does what it asks. On the shared questions, 11.6 points of quotations the model would have left unsourced receive a source, but the source is correct ($+5.8$) and wrong ($+6.1$) in equal measure. Asking an open-weight model to cite therefore adds sources to its quotations without making them reli-



(a) The cost of asking by textual fidelity



(b) The cost of asking by attribution

Figure 2: Decomposition of the pooled accuracy change from the implicit to the explicit prompt, by (a) textual fidelity and (b) attributional accuracy. Selection represents citations under shared questions. Quality represents the first shared citations. Volume represents explicit-only citations beyond the implicit count. Significance is computed using a question-level bootstrap with 8,000 resamples, each drawing the 500 questions with replacement and recomputing the delta of explicit minus implicit accuracy. Resampling is per question, as citations within an answer are not independent. The 95% confidence interval represents the central 95% of the computed deltas, and a result is considered significant when it excludes zero.

able.

The frontier models show the opposite pattern. Over the whole decomposition, 54.9 points of unsourced quotations become 53.9 points of correctly attributed ones, with misattribution rising by only 1.2, and almost all of this comes from quality (-51.5 unsourced, $+50.7$ correct); selection and volume each move fewer than three points. All attribution deltas above 5 points have 95% intervals that exclude zero.

The cost of asking is therefore paid only by the open-weight models, and paid on the citations the instruction forces rather than on the ones the models would volunteer. The per-model deltas are shown in Appendix 6.

6 Conclusion and Future Work

In this paper, we present a fine-grained taxonomy of citation errors in LLM-generated Islamic text that decouples a quotation's textual fidelity from its attributional accuracy. The taxonomy categories are grounded directly in expert adjudication notes. We evaluate twelve models under both explicit and implicit citation prompts, and extend the evaluation to eight more recent models under the implicit prompt. We analyse how prompting for evidence alters model generations. We show that this taxonomy captures errors that binary labels hide. The two model tiers also fail in different ways. Open-weight LLMs fail mostly by fabricating quotations, while frontier models tend to alter the original wording. Because models routinely attach plausible sources to fabricated text, a seemingly correct attribution does not guarantee the quote is genuine. Finally, the cost of prompting for citations falls almost entirely on the citations the model is forced to make rather than those it provides voluntarily. Future work can expand the taxonomy to distin-

guish fabricated sources and more specific forms of corruption, including omission, addition, conflation, and, for Hadith, meaning-changing alteration. Researchers can also study whether a hallucinated citation signals an overall unreliable answer. Models' tendency to append authoritative sources to entirely fabricated scripture poses severe risks in high-stakes settings. We hope the released annotations help benchmark, detect, and mitigate domain-specific hallucinations.

Limitations

The Hadith Matn rule evaluates whether the text matches an attested wording. It does not judge whether altered phrasing might be acceptable as transmission by meaning (*riwāya bi-l-ma'nā*). The reported corruption rates therefore reflect exact textual fidelity under this specific rule. The annotator model, Gemini Pro 3.1, is also one of the evaluated models. It assigned *corrupted* labels to 12% of the Hadith citations generated by its own family, and its agreement with the expert reference was measured on citations drawn from all twelve models, but we did not test whether it judges its own outputs more leniently than those of other models (Panickssery et al., 2024). Prompts expose the citation text without generator identity, and the correction check was run with a model from a different family than the annotator, providing cross-family evidence on the fidelity boundary.

Ethical Considerations

The released dataset includes model-generated citations, many of which are corrupted or fabricated Quranic and Hadith texts. This dataset is intended solely for evaluating hallucination detection and must not be used as a source of religious text. While we validated fine-grained labels against expert judgments on a sample and checked them against canonical corpora, an LLM annotator produced them, and they may contain errors. The per-model results report the fidelity of the citation and do not reflect the general reliability of a model on religious questions and should not be used to select a model for religious guidance. Answers should always be subject to qualified human review. The underlying data are used with permission from (Alharbi et al., 2026).

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A Prompts

A.1 Answer Generation

Explicit condition

أجب عن السؤال التالي و استشهد بآيات من القرآن الكريم و احاديث شريفة عند الاستشهاد بآية قرآنية، اذكر اسم السورة ورقم الآية. عند الاستشهاد بحديث، اذكر المصدر (البخاري، مسلم ، إلخ)

Answer the following question and cite verses from the Holy Quran and noble Hadith. When citing a Quranic verse, state the name of the Surah and the verse number. When citing a Hadith, state the source (al-Bukhari, Muslim, etc.)

Implicit condition

أجب عن السؤال التالي.

Answer the following question.

A.2 Annotator Prompts

Figures 3 and 4 show the annotator prompts for each citation type as deployed.

B Taxonomy Details

Table 3 lists the eight failure modes as the label permutations that define them. The three groups follow the textual-fidelity axis (correct, corrupted, and fabricated), and the rows within each group follow the attributional axis. We assign a mode using the procedure in Figure 1: we judge the quoted text first, judge attribution only when text exists to judge it against, and record components the model did not supply as unsupplied rather than incorrect, yielding the separate unsourced modes. Under *fabricated* there is no text to attribute, so the two fabrication modes differ only in whether an attribution was supplied.

Table 5 shows how we derived the fine-grained ground truth from the experts' free-text notes (Section 4), with one fabricated and one corrupted example. Each row shows one annotator's binary verdict, the note they attached, and the fine-grained label the note yields: a note stating the citation has no basis gives *fabricated*, while a note supplying the attested wording or naming the altered phrase gives *corrupted*. The three annotators' notes were read independently, and we take the majority label. Table 4 shows labelled examples of the modes from the released dataset.

C Additional Results

Table 6 shows the per-model decomposition of the effect of the cost of asking.

Annotator prompt for Quranic citation

Evaluate Arabic Quranic citation fragments for TEXTUAL FIDELITY and EXISTENCE. Write all step-by-step reasoning in English. Keep every quoted Arabic text in Arabic.
WHAT YOU ARE JUDGING. You judge whether the cited text is the real text of a Quranic verse that EXISTS — and how faithfully it is reproduced. The Quran has ONE canonical text and no riwaya variation, so the reference is the exact rasm of the verse. You are detecting whether a language model reproduced a real verse faithfully, altered it, or INVENTED something that is not Quran.
STEP 1 AYAH TEXT: classify into exactly one of three categories.

- **correct** = exact Quranic text. The rasm (consonantal skeleton) must be exact. Tashkil rule: if the citation has NO tashkil (no diacritics), a correct rasm alone is CORRECT; if the citation HAS tashkil, the tashkil must ALSO be correct. A partial quote is correct ONLY if it is CONTIGUOUS — a clean continuous run of the verse with nothing removed from inside it (for example, a clean prefix that stops at a word boundary).
- **corrupted** = a REAL verse exists and is still recoverable/identifiable, but the rendering is wrong in some way: a MIDDLE OMISSION (words dropped from inside the quote, whether or not marked by an ellipsis or dots), a substituted or wrong word, a wrong word in the middle, WRONG TASHKIL when tashkil is present, or a cut-off/incomplete word. You can still tell which real verse it is meant to be.
- **fabricated** = NOT a Quranic verse at all: invented text that cannot be recovered to any real verse in the Quran.

Key distinction: corrupted is RECOVERABLE (you can identify the intended real verse); fabricated is invented text, matching no real verse. A contiguous partial is correct; a partial with a middle omission is corrupted. In your reason, name the SPECIFIC omitted, substituted, or mis-voweled segment. If it is not Quran at all, state that it is fabricated (exists nowhere).
STEP 2 GATE (existence-based). Ayah fabricated it is not a real verse: set `source_label = "N/A"` and stop judging source. Ayah corrupted OR correct if the verse EXISTS: JUDGE source normally.
STEP 3 SOURCE (judge unless Ayah fabricated). Judge the reference ONLY against what is actually claimed. Never mark source incorrect for information that was not provided. A missing Ayah number is never an error. Judge only what is claimed.

- **none** = no reference given.
- **correct** = surah is right (whether or not a number is given), and if a number is given it is also right.
- **incorrect** = surah is wrong, or a given number is wrong.

STEP 4 CONFIDENCE (after reasoning). `factual_confidence` (High/Medium/Low): certainty in the judgment.
OUTPUT. Return ONLY this JSON object, with no prose outside it:

```
{
  "type": "Ayah",
  "ayah_label": "correct|corrupted|fabricated",
  "source_label": "correct|incorrect|N/A|none",
  "dropped_changes_meaning": true|false,
  "extraction_confidence": "High|Medium|Low",
  "factual_confidence": "High|Medium|Low",
  "reason": "<English; name the specific
   omitted/substituted/mis-voweled segment,
   or state it exists nowhere>"
}
```

Figure 3: The annotator prompt for Quranic citations.

Annotator prompt for Hadith citation

Evaluate Arabic Hadith citation fragments for WORDING FIDELITY and EXISTENCE. Write all step-by-step reasoning in English. Keep every quoted Arabic text in Arabic.
WHAT YOU ARE JUDGING. You judge whether the cited text is the real wording of a Hadith that EXISTS — NOT its authenticity grade. "Existence" means the Hadith is documented somewhere in the corpus (any collection, even if the Hadith is weak or forged). You are detecting whether a language model reproduced a real text faithfully, altered it, or INVENTED something that does not exist.
STEP 1 MATN: classify into exactly one of three categories.

- **correct** = the wording matches the exact lafz of a Hadith that EXISTS in the corpus (in any authentic riwaya / transmission). A clean prefix (partial but exact, stopping at a word or phrase boundary and dropping NOTHING from the middle) is still correct. The wording is NOT correct if it changes the substance: an added, dropped, or substituted word or clause, or a conflation of two Hadiths.
- **corrupted** = a REAL Hadith exists, but THIS wording changes its substance in a way above (an added, dropped, or substituted word or clause, or a conflation). The underlying Hadith EXISTS; only the wording is altered.
- **fabricated** = NO such Hadith exists ANYWHERE in the corpus (invented text).

CRITICAL mawdu (forged) or daif (weak) is NOT fabricated. A Hadith that scholars documented as forged or weak STILL EXISTS in the literature. If the cited text matches such a documented Hadith, it is "correct" or "corrupted" by the wording rule above — never "fabricated". Reserve "fabricated" for text invented by the model that is found in NO book and NO collection. In your reason, name the SPECIFIC altered, added, dropped, or conflated word or clause (do not just say "corrupted"). If fabricated, state that it exists nowhere.
STEP 2 HADITH GRADE (record, do not use it for the label). Independently of STEP 1, state the authenticity grade of the underlying Hadith if one exists: one of `"sahih" | "hasan" | "daif" | "mawdu" | "unknown" | "none"` (`"none"` only when fabricated). This is CAPTURED for downstream use; it must NOT change `Matn_label`. A mawdu Hadith with exact wording is still `Matn_label="correct"`, `Hadith_grade="mawdu"`.
STEP 3 GATE (existence-based). Matn fabricated then no real chain or source exists: set `Isnad_label="N/A"` and `source_label="N/A"`. Matn corrupted OR correct the Hadith EXISTS: JUDGE Isnad and source normally.
STEP 4 ISNAD (judge unless Matn fabricated). Judge whether the CLAIMED narrator matches the TRUE narrator of this Hadith. The relevant narrator is the COMPANION (al-sahabi / al-rawi) the Hadith is attributed to.

- **correct** = the claimed Companion is the correct narrator of this Hadith.
- **incorrect** = a DIFFERENT Companion actually narrates it (a genuine narrator conflict), even if the cited chain contains otherwise-real transmitters. Resolve aliases (Ibn Umar = Abdullah ibn Umar).
- **none** = no chain or narrator is cited at all.

Do NOT judge chain-grade or transmitter validity here only Companion identity. In your reason, name the Companion you determined to be the true narrator. If you judge the claimed Companion correct, say briefly in which narration that Companion narrates this wording.
STEP 5 SOURCE (judge unless Matn fabricated). Does the CLAIMED book actually contain THIS Hadith?

- **incorrect** = a real Hadith attributed to the WRONG book. Unqualified "رواه البخاري" means SAHIH al-Bukhari specifically; a Hadith only in al-Adab al-Mufrad cited as "al-Bukhari" is an overclaim = incorrect. "متفق عليه" requires presence in BOTH Sahihs.
- **none** = no source claimed.

STEP 6 CONFIDENCE (after reasoning). `factual_confidence` (High/Medium/Low): your certainty in the factual judgment.
OUTPUT. Return ONLY this JSON object, with no prose outside it:

```
{
  "type": "Hadith",
  "Matn_label": "correct|corrupted|fabricated",
  "Hadith_grade": "sahih|hasan|daif|mawdu|unknown|none",
  "Isnad_label": "correct|incorrect|N/A|none",
  "source_label": "correct|incorrect|N/A|none",
  "factual_confidence": "High|Medium|Low",
  "reason": "<English; name the specific altered/added/
   dropped/conflated element, the true narrator if Isnad
   is wrong, and the correct book if source is wrong;
   or state it exists nowhere>"
}
```

Figure 4: The annotator prompt for Hadith citations.

Failure mode	Text	Isnad	Claimed source	Definition
Correct Citation	correct	correct	correct	Correct text, with all attributions correct.
Misattribution	correct	**incorrect**	**incorrect**	Correct text, with an incorrect attribution.
Unsourced Quotation	correct	—	—	Correct text, with no attribution supplied.
Textual Corruption	corrupted	correct	correct	Altered text, with a correct attribution.
Corrupted Misattribution	corrupted	**incorrect**	**incorrect**	Altered text, with an incorrect attribution.
Unsourced Corrupted Quotation	corrupted	—	—	Altered text, with no attribution supplied.
Attributed Fabrication	fabricated	supplied, not judged		Non-existent text, with an attribution supplied.
Bare Fabrication	fabricated	—	—	Non-existent text, with no attribution supplied.

Table 3: The fine-grained taxonomy of Islamic citation errors, stated as the label permutations that define each mode. — marks a component the model did not supply, which is not counted as a failure.

		Attribution	
Failure mode	**Quoted text**	**Isnad**	**Claimed source**
Correct Citation Hadith	من غشنا فليس منا *man ghashshanā fa-laysa minnā* "He who acted dishonestly towards us is not of us."	أبو هريرة Abū Hurayrah	رواه مسلم Muslim
Misattribution Quran	وعاشروهن بالمعروف *wa-ʿāshirūhunna bi-l-maʿrūf* "And live with them in kindness."	—	سورة الطلاق (٢) al-Ṭalāq 65:2
Textual Corruption Quran	وَلَا تَعْضُلُوهُنَّ *wa-lā taʿḍulūhunna* "Do not prevent them."	—	سورة البقرة، الآية ٢٣٢ al-Baqarah 2:232
Corrupted Misattribution Quran	وَاذْكُرْ فِي الْكِتَابِ إِدْرِيسَ إِنَّه كَانَ خَاشِعًا مُنِيبًا *wa-dhkur fī al-kitābi Idrīsa innahu kāna khāshiʿan munībban* "And mention in the Book, Idrees. Indeed, he was a man of truth and a prophet."	—	مريم : ١٣٩ Maryam 19:139
Attributed Fabrication Hadith	أذّنوا إذا أردتم *adhdhinū idhā aradtum* Lit. "Call the adhan when you wish."	—	البخاري، برقم ٦٧٠ al-Bukhārī, no. 670

Table 4: Examples from the released annotated data, covering a subset of the failure modes. The quoted text is coloured by textual fidelity (green correct, yellow corrupted, red fabricated), with altered words underlined in the transliteration. Attribution is coloured green when correct, red when incorrect, and grey when supplied with fabricated text and not judged; — marks a component not supplied or not applicable.

Binary Label	Expert Notes	Derived Label
Cited Matn: اللهمَ اجعلْهُ في رِجزٍ، واجعلْهُ من أَقلَ أُمَتي عَدَدًا		
incorrect	هذا ليس حديثًا. *hādhā laysa áadīthan.* "This is not a Hadith."	fabricated
incorrect	ليس حديثا *laysa áadīthan* "Not a Hadith."	fabricated
incorrect	حديث لا أصل له *áadīth lā aála lah* "A Hadith with no basis."	fabricated
Cited Matn: ما من الأنبياء نبي إلا أُعطي من الآيات ما مثله آمن عليه البشر…		
correct	المرجع: رقم الحديث الصحيح: (٤٩٨١). *al-marji: raqm al-áadīth al-áaáīá: (4981)* "The reference: the correct Hadith number is (4981)."	correct
incorrect	ما مِنَ الأنبياءِ نَبيٌ إلّاْ أُعطيَ ما مِثلُه آمَنَ عليه البَشَرُ…	corrupted
incorrect	الخطأ في قوله أُعطي من الآيات ما مثله *al-khaáa fī qawlihi "uáiya mina l-āyāti mā mithluh"* "The error is in the phrase 'given of the signs the like of which'."	corrupted

Table 5: Deriving fine-grained labels from the expert notes: one fabricated and one corrupted example with the three annotators' notes. Notes stating the text has no basis give *fabricated*; notes supplying the attested wording or naming the altered phrase give *corrupted*; correct labels are carried over unchanged, and the majority over the derived labels is taken.

Model	Δ Total (pp)	Δ Selection (pp)	Δ Quality (pp)	Δ Volume (pp)
Llama-3.1-8B	**−25.9** [−35.6, −16.7]	**−7.9** [−14.3, −2.7]	**−10.7** [−17.7, −3.8]	**−7.4** [−11.6, −3.6]
Phi-3.5-mini	**−18.7** [−31.1, −9.2]	**−12.8** [−21.1, −5.2]	+0.1 [−10.3, 9.4]	**−6.0** [−12.5, −0.8]
Gemma-3-12B	**−12.5** [−20.3, −5.7]	−1.9 [−5.2, 1.5]	−6.2 [−13.1, 0.0]	−4.4 [−9.2, 0.1]
AceGPT-v2-8B	−9.0 [−22.2, 3.4]	**−18.3** [−32.9, −4.3]	**+18.4** [2.6, 35.0]	**−9.1** [−19.0, −2.5]
Qwen3-8B	**−8.8** [−12.5, −5.3]	**−2.6** [−5.1, −0.2]	−2.7 [−6.8, 1.3]	**−3.4** [−6.2, −0.9]
Command-R-7B	−8.0 [−18.6, 1.3]	−1.7 [−9.1, 5.5]	+2.9 [−7.6, 13.8]	**−9.2** [−15.5, −2.8]
Mistral-7B	**−7.2** [−13.8, −1.8]	**−6.9** [−12.7, −2.2]	+4.0 [−2.2, 11.0]	**−4.2** [−9.8, −0.2]
ALLaM-7B	−1.9 [−7.2, 3.5]	−1.0 [−4.6, 2.6]	+3.5 [−3.1, 10.1]	**−4.5** [−8.0, −0.9]
Fanar-1-9B	+0.5 [−5.2, 6.1]	−0.2 [−4.1, 3.7]	+3.7 [−3.1, 10.5]	−3.1 [−6.8, 0.6]
GPT-5.4†	+2.5 [−0.1, 5.3]	+1.1 [−0.8, 3.2]	+0.7 [−2.6, 4.2]	+0.8 [−1.5, 3.1]
Gemini3.1-Pro†	**+3.8** [1.0, 6.9]	+1.6 [−0.3, 4.0]	**+4.1** [1.4, 7.0]	**−2.0** [−3.4, −0.5]
Jais-13B	+17.3 [−5.7, 37.4]	−7.7 [−28.9, 13.2]	**+25.0** [5.9, 44.4]	0.0 [−13.3, 12.1]

Table 6: Per-model decomposition of the accuracy change from the implicit to the explicit prompt. Selection represents citations under shared questions. Quality represents first shared citations. Volume represents explicit-only citations beyond the implicit count. Values report estimated changes with 95% confidence intervals in brackets. Bold estimates indicate intervals that exclude zero. †marks frontier models.

		Correct text				Corrupted text				Fabricated		
Model	**N**	**All**	**Correct**	**Misattr.**	**Unsrc.**	**All**	**Corrupt.**	**Misattr.**	**Unsrc.**	**All**	**Attrib.**	**Bare**
Gemini3.1-Pro†	2,186	**94.6%**	92.7%	1.0%	1.0%	**5.4%**	5.2%	0.2%	0.0%	**0.0%**	0.0%	0.0%
GPT-5.4†	2,816	**94.2%**	91.5%	1.6%	1.0%	**5.8%**	5.5%	0.2%	0.1%	**0.0%**	0.0%	0.0%
ALLaM-7B	1,693	**78.2%**	60.3%	17.2%	0.6%	**16.0%**	10.6%	5.1%	0.4%	**5.8%**	5.7%	0.1%
Fanar-1-9B	1,151	**67.3%**	41.0%	24.0%	2.3%	**15.6%**	7.8%	7.5%	0.3%	**17.1%**	16.6%	0.5%
Jais-13B	222	**49.1%**	32.4%	5.0%	11.7%	**26.1%**	17.6%	5.0%	3.6%	**24.8%**	23.0%	1.8%
AceGPT-v2-8B	325	**27.7%**	16.9%	7.7%	3.1%	**24.0%**	11.1%	9.8%	3.1%	**48.3%**	46.5%	1.8%
Command-R-7B	1,143	**18.7%**	10.6%	7.1%	1.0%	**19.7%**	7.8%	11.5%	0.3%	**61.6%**	61.1%	0.5%
Gemma-3-12B	1,922	**10.6%**	6.8%	3.4%	0.4%	**22.7%**	13.6%	8.6%	0.5%	**66.8%**	66.5%	0.2%
Qwen3-8B	1,547	**9.0%**	2.5%	5.0%	1.5%	**10.0%**	2.7%	6.9%	0.5%	**81.0%**	79.8%	1.2%
Llama-3.1-8B	1,340	**3.7%**	0.9%	2.0%	0.8%	**7.3%**	2.5%	4.7%	0.1%	**89.0%**	88.1%	0.8%
Phi-3.5-mini	876	**2.6%**	0.5%	1.6%	0.6%	**0.6%**	0.0%	0.5%	0.1%	**96.8%**	96.0%	0.8%
Mistral-7B	1,308	**2.2%**	0.4%	1.4%	0.5%	**2.7%**	0.9%	1.7%	0.1%	**95.1%**	92.6%	2.5%
Frontier (2)	5,002	**94.4%**	92.0%	1.3%	1.0%	**5.6%**	5.3%	0.2%	0.1%	**0.0%**	0.0%	0.0%
Open-weight (10)	11,527	**25.6%**	16.8%	7.7%	1.2%	**13.4%**	6.8%	6.1%	0.4%	**61.0%**	60.1%	0.9%

Table 7: Per-model distribution over the eight terminal classes of the citation taxonomy under the explicit prompt. †marks frontier models.